

$$330. \text{ Given exp.} = \frac{(a+b)^2 + (a-b)^2}{a^2 + b^2} \text{ (where } a = 856, b = 167\text{)}$$

$$= \frac{2(a^2 + b^2)}{(a^2 + b^2)} = 2.$$

$$331. \text{ Given exp.} = \frac{(a+b)^2 - (a-b)^2}{ab} = \frac{4ab}{ab} \\ = 4 \text{ (where } a = 469, b = 174\text{)}.$$

$$332. 2ab = (a^2 + b^2) - (a - b)^2 = 29 - 9 = 20 \Rightarrow ab = 10.$$

$$333. \frac{x^2 - 1}{x + 1} = 4 \Leftrightarrow \frac{(x+1)(x-1)}{x+1} = 4 \Leftrightarrow x-1 = 4 \Leftrightarrow x = 5.$$

$$334. \text{ If } a = 3\frac{2}{3}, b = 2\frac{1}{2}, c = 4\frac{3}{4}, d = 3\frac{1}{3}, \text{ then}$$

$$\text{Given exp.} = \frac{(a^2 - b^2)}{(c^2 - d^2)} \div \frac{(a-b)}{(c-d)} = \frac{(a^2 - b^2)}{(c^2 - d^2)} \times \frac{(c-d)}{(a-b)} \\ = \frac{(a+b)}{(c+d)} = \frac{3\frac{2}{3} + 2\frac{1}{2}}{4\frac{3}{4} + 3\frac{1}{3}} = \frac{\frac{11}{3} + \frac{5}{2}}{\frac{19}{4} + \frac{10}{3}} \\ = \frac{\frac{37}{6} \times \frac{12}{97}}{\frac{74}{97}}.$$

$$335. \text{ Given exp.} = \frac{a^2 - b^2}{a + b} = a - b \\ = \left(1 + \frac{1}{1 + \frac{1}{100}}\right) - \left(1 - \frac{1}{1 + \frac{1}{100}}\right) \\ = 2 \times \frac{1}{(101/100)} = 2 \times \frac{100}{101} = \frac{200}{101}.$$

$$336. \text{ Given exp.} = \frac{769435 \times 770001 - (770001 - 769435)}{769435 + 770001 \times (769435 - 1)} \\ = \frac{769435 \times 770001 - 770001 + 769435}{769435 + 770001 \times 769435 - 770001} = 1.$$

$$337. a - b = 1 \Rightarrow (a - b)^3 = 1^3 = 1 \Rightarrow a^3 - b^3 - 3ab(a - b) = 1 \\ \Rightarrow a^3 - b^3 - 3ab = 1 [\because a - b = 1]$$

$$338. x - y = 1 \Rightarrow (x - y)^2 = 1 \Rightarrow x^2 + y^2 - 2xy = 1 \\ \Rightarrow 41 - 2xy = 1 \Rightarrow 2xy = 40. \\ \therefore (x + y)^2 = x^2 + y^2 + 2xy = 41 + 40 = 81 \\ \Rightarrow (x + y) = \pm 9.$$

$$339. x + y = 1 \Rightarrow (x + y)^2 = 1^2 = 1 \Rightarrow x^2 + y^2 + 2xy = 1 \\ \Rightarrow x^2 + y^2 = 1 - 2xy \\ \Rightarrow (x^2 + y^2)^2 = (1 - 2xy)^2 \\ \Rightarrow x^4 + y^4 + 2x^2y^2 = 1 + 4x^2y^2 - 4xy \\ \Rightarrow 2x^2y^2 - 4xy = 17 - 1 = 16 [\because x^4 + y^4 = 17] \\ \Rightarrow x^2y^2 - 2xy = 8.$$

$$340. \frac{x^2 + y^2 + z^2 - yz - zx - xy}{a^2 + b^2 + c^2 - ab - bc - ca} \\ = \frac{(a+m)^2 + (b+m)^2 + (c+m)^2 - (b+m)(c+m) - (c+m)(a+m) - (a+m)(b+m)}{a^2 + b^2 + c^2 - ab - bc - ca}$$

$$\frac{a^2 + m^2 + 2am + b^2 + m^2 + 2bm + c^2 + m^2 + 2cm - bc - bm - cm - m^2 - ca - cm - am - m^2 - ab - am - bm - m^2}{a^2 + b^2 + c^2 - ab - bc - ca} \\ = \frac{a^2 + b^2 + c^2 - ab - bc - ca}{a^2 + b^2 + c^2 - ab - bc - ca} = 1.$$

$$341. (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) \\ \Rightarrow 2(ab + bc + ca) = (a + b + c)^2 - (a^2 + b^2 + c^2) \\ = 169 - 69 = 100$$

$$\Rightarrow ab + bc + ca = 50.$$

$$342. \text{ Given: } x^2 + y^2 + z^2 - 64 = -2(xy - yz - zx) \quad \dots(i)$$

$$\text{Now, } [x + y + (-z)]^2 = x^2 + y^2 + z^2 + 2(xy - yz - zx)$$

$$\Rightarrow (3z - z)^2 = x^2 + y^2 + z^2 + 2(xy - yz - zx)$$

$$\Rightarrow -2(xy - yz - zx) = (x^2 + y^2 + z^2) - (2z)^2 \quad \dots(ii)$$

$$\text{From (i) and (ii), we get: } (2z)^2 = 64 \Leftrightarrow 4z^2 = 64$$

$$\Leftrightarrow z^2 = 16$$

$$\Leftrightarrow z = 4.$$

$$343. \text{ Given exp.} = \left(\frac{a^3 + b^3}{a^2 + b^2 - ab}\right) = (a + b) \text{ (where } a = 785, b = 435\text{)} \\ = (785 + 435) = 1220.$$

$$344. \text{ Given exp.} = \left(\frac{a^2 + ab + b^2}{a^3 - b^3}\right) = \left(\frac{1}{a - b}\right) \\ \text{(where } a = 147, b = 143\text{)} = \left(\frac{1}{147 - 143}\right) = \frac{1}{4}.$$

$$345. \text{ Let } \frac{13^3 + 7^3}{13^2 + 7^2 - x} = 20.$$

$$\text{Then, } \frac{13^3 + 7^3}{13 + 7} = 13^2 + 7^2 - x$$

$$\Leftrightarrow 13^2 + 7^2 - 13 \times 7 = 13^2 + 7^2 - x$$

$$\Leftrightarrow x = 13 \times 7 = 91.$$

$$346. \text{ Given exp.} = \frac{a^3 - b^3}{a^2 - b^2} = \frac{(a-b)(a^2 + ab + b^2)}{(a-b)(a+b)} = \frac{(a^2 + ab + b^2)}{(a+b)}$$

$$= \frac{\left(\frac{3}{5}\right)^2 + \left(\frac{3}{5} \times \frac{2}{5}\right) + \left(\frac{2}{5}\right)^2}{\left(\frac{3}{5} + \frac{2}{5}\right)}$$

$$= \frac{\frac{9}{25} + \frac{6}{25} + \frac{4}{25}}{\frac{5}{5}} = \frac{19}{25}.$$

$$347. \text{ Given exp.} = \frac{a^3 + b^3 + c^3 - 3abc}{a^2 + b^2 + c^2 - ab - bc - ca} \\ = a + b + c = (38 + 34 + 28) = 100.$$

$$348. \text{ Since } (x - y) + (y - z) + (z - x) = 0, \\ \text{so } (x - y)^3 + (y - z)^3 + (z - x)^3 = 3(x - y)(y - z)(z - x).$$

$$\therefore \text{ Given exp.} = \frac{3(x - y)(y - z)(z - x)}{9(x - y)(y - z)(z - x)} = \frac{1}{3}.$$